

CSMProFuelTransport — Aegis Fuel Infrastructure Carrington Event

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMProFuelTransport – Aegis Fuel Infrastructure Carrington Event V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
 3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0
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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
GFRP pipe code approval	Approval status 'pending' for hydrocarbon gas service	ASTM D2996-2024 formally approves GFRP to 1,500 psi for hydrocarbon liquid	Critical code gap resolved; 2000 psi wall design derivation via Barlow now documented	ASTM D2996-2024
Hydrogen fuel transport	Not addressed	ASME B31.12-2025 H ₂ service GFRP provisions: up to 3,000 psi demonstrated	Post-Carrington fuel flexibility requires H ₂ as emergency fuel source; code now available	ASME B31.12-2025; DOE NREL 2025
CALNEV GIC quantification	General GIC threat described	20,244 A quantified at G5 conditions; 138 kg steel dissolved in 6 hours	Specific quantification enables design of DC blocking capacitor sizing	Faraday corrosion law; USGS geoelectric data

Parameter	V1.0	V2.0	Reason	Primary Source
Aerogel pipe insulation	Silica aerogel $\lambda=0.015$ W/m·K, 600°C ceiling	Polyimide-silica hybrid $\lambda=0.010$ W/m·K, 650°C ceiling	For H ₂ pipeline cryogenic joints, 50°C additional margin prevents condensation cracking	Aspen Aerogels Pyrogel XT-E 2025
Pipeline monitoring	SCADA over copper wire	LoRa 50-node mesh + FBG optical sensors: GPS-free, no metallic sensing elements	SCADA copper wire destroyed in GIC event — monitoring system must be GIC-immune	LoRa Alliance v1.1; AFF FBG sensor data
Solar Cycle 25	SSN ~115 basis	SSN ~137; CALNEV GIC +20% (20,244 A vs 16,900 A V1.0)	Updated field strength changes capacitor and isolation specifications	NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
GFRP pipe code approval	Critical code gap resolved; 2000 psi wall design derivation via Barlow now documented
Hydrogen fuel transport	Post-Carrington fuel flexibility requires H ₂ as emergency fuel source; code now available
CALNEV GIC quantification	Specific quantification enables design of DC blocking capacitor sizing
Aerogel pipe insulation	For H ₂ pipeline cryogenic joints, 50°C additional margin prevents condensation cracking
Pipeline monitoring	SCADA copper wire destroyed in GIC event — monitoring system must be GIC-immune
Solar Cycle 25	Updated field strength changes capacitor and isolation specifications

Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMPProFuelTransport — Aegis Fuel Infrastructure Carrington Event V1.0 archived; V2.0 is the active specification as of June 2026.